

SatsPath v2: DNSSEC Delegation

Overview

To provide strong, cryptographic binding between a domain-based identifier (e.g., `alice@example.com`) and its authoritative resolving server, SatsPath uses DNSSEC. This document specifies how a namespace owner configures their DNS to prove authority to SatsPath resolvers.

The DNS Record

The descriptor is published as a TXT record at `_satspath.<domain>`. For `example.com`, the record must be at `_satspath.example.com`.

Format

The content is a string containing semicolon-separated key-value pairs.

```
v=sp2; e=https://satspath.example.com; k=xpub...; l=log_id...; w=witness_policy;
```

Fields

- **v** (required): Protocol version. MUST be `sp2`.
- **e** (required): Authoritative S2S endpoint URL. MUST be HTTPS.
- **k** (required): Hex-encoded compressed secp256k1 public key of the namespace authority.
- **l** (required): The `log_id` (Merkle root hash or ID) of the transparency log.
- **w** (optional): Witness policy identifier.

Validation Rules

1. **DNSSEC Required:** The resolver MUST validate the record using DNSSEC from a configured local trust anchor. Responses without the `AD` bit proven locally MUST NOT be treated as `domain_verified`.
2. **TTL Enforcement:** Resolvers MUST cache the record according to its TTL and MUST NOT use stale records.
3. **No Silently Falling Back:** If DNSSEC verification fails or the signature is bogus, the resolver MUST fail closed (error) rather than falling back to an unverified state.